Answer key - Density and Pressure

GCE Advanced Level - Physics - Corrige

Detailed method for every exercise. Follow the substitutions line by line.
Corrige detaille de chaque exercice. Suis les substitutions ligne par ligne.

1. Mass 6 kg, volume 2 m^3. Density (kg/m^3, number only)?
Masse 6 kg, volume 2 m^3. Masse volumique (kg/m^3, nombre seul) ?

Answer: 3
? = 6 / 2 = 3 kg/m^3

2. Wood (? ~ 600 kg/m^3) placed in water (? = 1000 kg/m^3) will...
Du bois (? ~ 600 kg/m^3) placé dans l'eau (? = 1000 kg/m^3)...

- Float / Flotte *
- Sink / Coule
- Stay suspended / Reste en suspension

Answer: Float
Less dense than water -> floats.
Moins dense que l'eau -> flotte.

3. Force 100 N acts on area 2 m^2. Pressure (Pa, number only)?
Force 100 N sur surface 2 m^2. Pression (Pa, nombre seul) ?

Answer: 50
P = 100 / 2 = 50 Pa

4. A crate exerts 300 N over area 0.5 m^2. Pressure (Pa, number only)?
Une caisse exerce 300 N sur 0,5 m^2. Pression (Pa, nombre seul) ?

Answer: 600
300 / 0.5 = 600 Pa
300 / 0,5 = 600 Pa

5. Why does a sharp knife cut better than a blunt one?
Pourquoi un couteau tranchant coupe-t-il mieux qu'un couteau émoussé ?

- Smaller area gives higher pressure / Surface plus petite -> pression plus forte *
- Larger area gives higher pressure / Surface plus grande -> pression plus forte
- It has more force / Il exerce plus de force
- It weighs more / Il est plus lourd

Answer: Smaller area gives higher pressure
Smaller area -> larger pressure for same force.
Plus petite surface -> plus grande pression pour même force.

6. Mass 10 kg, volume 5 m^3. Density (kg/m^3)?
Masse 10 kg, volume 5 m^3. Masse volumique (kg/m^3) ?

Answer: 2
2 kg/m^3

7. An object with density less than water will...
Un objet moins dense que l'eau...

Mega Mighty Sixers - contenu original, diffusion libre aux membres de l association

- Float / Flotte *
- Sink / Coule

Answer: Float
Floats.
Flotte.

8. The SI unit of pressure is the...
L'unité SI de la pression est...

- Pascal (Pa) *
- Newton (N)
- Joule (J)

Answer: Pascal (Pa)
Pascal (Pa).

9. Force 400 N over 0.1 m^2. Pressure (Pa)?
Force 400 N sur 0,1 m^2. Pression (Pa) ?

Answer: 4000
4000 Pa

10. Snowshoes prevent sinking because they...
Les raquettes empêchent l'enfoncement car elles...

- Increase area, reduce pressure / Augmentent surface, réduisent pression *
- Decrease area, increase pressure / Diminuent surface, augmentent pression
- Reduce weight / Réduisent le poids

Answer: Increase area, reduce pressure
Increase area -> decrease pressure.
Augmentent la surface -> diminuent la pression.

Mega Mighty Sixers - contenu original, diffusion libre aux membres de l association